

Discussion

Developing a controlled vocabulary

- Three axes:
 - Topics: What is the paper about? (i.e. diaspora, mobility, gender, ritual, etc)
 - Methods: What methods were applied in the paper? (i.e. spatial analysis, genetics, stable isotope analysis, finds analysis)
 - Kind of contribution (i.e. field report, “standard” scientific paper, review of prior work, opinion)
- The medium (i.e. journal article, book, chapter in an edited volume, etc) will be identified by CrossRef and does not need to be annotated.
- It’s ok to assign multiple terms to each record.

Coding process

- A research assistant will be able to start work in September, and will be able to work for four weeks.
- Limited to review of titles and abstracts, or if there is no abstract, the introductions and conclusions.
- Since this does not involve comprehensive or open coding, we can just do this in a spreadsheet rather than some specialized QDA software.
- The research assistant will ideally be a more experienced student in the field of Viking Age archaeology.
- Zack will schedule weekly check-ins with the research assistant and make himself available to address any methodological questions or concerns.

Analysis of references

- We won’t be able to qualitatively annotate references due to lack of time relative to the corpus’ volume.
- NLP using scite is not as useful as initially thought.
 - tends to “play it safe” by coding references as neutral mentions when human coders identify a more robust stance (see Bakker et al. 2023).
 - Its corpus is limited to works put out by partnered publishers.
- Potential alternative to scite, but requires greater knowledge of NLP to implement:
 - <https://github.com/davidjurgens/citation-function>
 - <https://jurgens.people.si.umich.edu/citation-function>
- We may be able to derive a sample based on network statistics, i.e. the most or least central references.

Analysis of gender

- Our intent is to identify citation biases against women.
- The goal is not to identify gender diversity, which is not supported by our method.

Predictions and expectations

- We’re gonna find sub-clusters within clusters, and bridging clusters.
- Some sub-clusters will be tight-knit and closed, others very broad and open.
- Some will be very focused on specific topics.
- There will be many links across institutions, they are not really that fixed.
- Some citation chains will emerge that reflect casual observations (just a few examples):
 - Lots of cross-references within the Mälaren valley community.
 - Tight-knit cluster among papers that are concerned with ritual.
 - Major computational cluster at Aarhus.

- If had a broader time horizon, we might be able to trace “genealogies of affiliation” by tracing how identities change based on the evolution of their collaborative networks.
- Clustering in methods by “academic upbringing”, which has a localized character
- We may be able to find associations based on age, but mainly by doing a rough assessment of the most and least central nodes, rather than keying age into the data.

Additional ideas and insights

- Keeping track of topics independently from citational links enables us to validate whether topics are actually clustered around certain individuals or papers, or whether there are diverse centres.
- We’ll be able to do some non-network descriptive statistics:
 - Distribution of topics by affiliations of authors.
 - Distribution of topics by number of co-authors.
 - Temporal distributions of genders of lead authors.
 - Gender of lead author by size and genders of their co-authors.

Logistics

- Keep things on GitHub for now.
- Keep our eyes open for alternative collaborative writing platforms that may emerge over the next few months (including via the archaeo-social collective).

Action items

Next meeting: Monday June 2, 15:15 CEST / 9:15 EST

Zack:

- Assemble a series of core variables to be included in a data frame
- Initiate a script to scrape abstracts and metadata
- Modify the research protocol with today’s notes
 - Expectations and predictions
 - Tone down expectations for analysis of references (based on volume of papers, limited time)
 - Document the controlled vocabulary and coding process

Julie:

- Draft controlled vocabulary for article annotation

Isak:

- Review and modify the R script

Links and references

- Peterson, David, and Aaron Panofsky. 2023. “Metascience as a Scientific Social Movement.” *Minerva* 61 (2): 147–74. <https://doi.org/10.1007/s11024-023-09490-3>.
- Bakker, Caitlin, Nicole Theis-Mahon, and Sarah Jane Brown. 2023. “Evaluating the Accuracy of Scite, a Smart Citation Index.” *Hypothesis: Research Journal for Health Information Professionals* 35 (2). <https://doi.org/10.18060/26528>.
- Thesis by Julie’s former student, about us/we/them in Viking Age literature.
- Open source alternative to scite:
 - <https://github.com/davidjurgens/citation-function>
 - <https://jurgens.people.si.umich.edu/citation-function>